

1. DATASET OVERVIEW & FILE STRUCTURE

This documentation covers the completed machine learning task for predicting residential asset valuations. The objective is to leverage a descriptive dataset containing features that influence transactional parameters to train an accurate regression model. The project utilizes two primary file subsets: **train.csv** (the structured training set containing both predictive features and historical target outcomes to optimize weights) and **test.csv** (the independent test split preserved to evaluate generalization performance).

2. TARGET VARIABLE ANALYSIS: The primary prediction target matches the variable `SalePrice`, which outlines the final closing value transaction for assets in currency units. As explicitly validated by the distribution curve, structural metrics show a significant positive/right skew, with a steep accumulation centered inside the entry-to-median brackets. The distribution features a truncated rise over entry-level thresholds and an elongated, volatile tail tracking toward custom luxury developments that surpass top percentiles. To stabilize variance constraints and minimize structural heteroscedasticity across ordinary least-squares configurations, a logarithmic mapping transformation ($\log_{10}$) is implemented directly over the target vector.

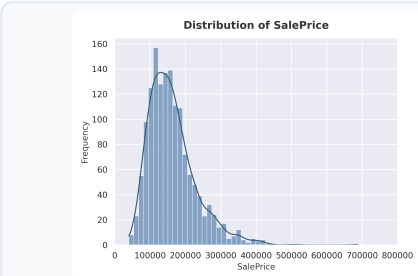


Figure 1.1: Target distribution histogram.

3. STRUCTURAL FEATURE GLOSSARY (CORE DIMENSIONS)

Feature Key	Category Type	Structural Context & Engineering Relevance
SalePrice	Continuous Target	The property's final transactional value. Objective function target variable.
MSSubClass	Nominal Categorical	Identifies specific type/age architectural dwelling configuration designs.
MSZoning	Nominal Categorical	General agricultural, commercial, industrial, or high/low-density residential zone classifications.
LotArea	Continuous Numerical	Total square footage of parcel area. Significant predictor for baseline ground valuation.
OverallQual	Ordinal Integer	Rates overall material and finish quality from 1 (Poor) to 10 (Very Excellent).
GrLivArea	Continuous Numerical	Total square footage of above-grade ground living area. High correlation with price.

4. ENCODED CATEGORICAL ATTRIBUTES EXAMPLES

MSSubClass Mappings:

20	1-STORY 1946 & NEWER ALL STYLES
30	1-STORY 1945 & OLDER
60	2-STORY 1946 & NEWER
90	DUPLEX - ALL STYLES AND AGES

MSZoning Mappings:

A	Agriculture	C	Commercial
FV	Floating Village Residential		
RL	Residential Low Density		
RM	Residential Medium Density		

1. SYNTHETIC FEATURE DERIVATION

- **Total Living Surface Footprint (TotalSF):**
Combines basement, 1st floor, and 2nd floor metrics into a unified surface proxy. This handles separate multicollinear fields and creates a strong, clean physical volume signal.
- **Cumulative Fractional Bathrooms (TotalBathrooms):**
Weights partial washrooms at 0.5 and aggregates them with full bathroom counts to create a linear continuous indicator reflecting accurate functional asset utility.
- **Total Exterior Porch Footprint (TotalPorchSF):**
Combines open, enclosed, three-season, and screened porch square footage. This eliminates sparse zero-variance null records across individual columns.

```
// Pseudocode: Feature Derivation Sequence
COMPUTE TotalSF = TotalBsmtSF + 1stFlrSF + 2ndFlrSF
COMPUTE TotalBathrooms = FullBath + (0.5 * HalfBath) + BsmtFullBath + (0.5 * BsmtHalfBath)
COMPUTE TotalPorchSF = OpenPorchSF + EnclosedPorch + 3SsnPorch + ScreenPorch
```

2. HIGH-SPARSITY DIMENSIONALITY PRUNING (FEATURE DROPPING)

Reasoning: Arbitrary indices like `Id` lack predictive value and can introduce unwanted patterns. High-sparsity elements like `Alley`, `PoolQC`, `Fence`, and `MiscFeature` contain over 90% missing data, making them statistical noise. Removing them ensures structural data integrity and reduces model variance.

3. COLUMN TRANSFORMATIONS: NUMERICAL IMPUTATION & CATEGORICAL ENCODING

- **Numerical Mean Imputation (SimpleImputer):**
Replaces missing values with the column average. This allows distance and gradient-based estimators to process rows seamlessly without shifting overall distribution traits.
- **First-Drop One-Hot Encoding (OneHotEncoder):**
Converts textual qualitative groups into binary columns so standard regression algorithms can process them. Dropping the first category flags out redundant bits and eliminates dummy variable traps.

4. VARIANCE STABILIZATION & ROBUST INTERQUARTILE OUTLIER FILTERS

- **Logarithmic Target Rescaling (log1p):**
Applies spatial scaling to shift highly skewed valuation variables into a balanced normal curve, uniforming variance profiles.
- **Interquartile Range (IQR) Boundary Truncation:**
Isolates the central 50% core data tier ($IQR = Q3 - Q1$). Profiles breaking outside bounds $[Q1 - 1.5 \cdot IQR, Q3 + 1.5 \cdot IQR]$ are filtered to stabilize regression paths.

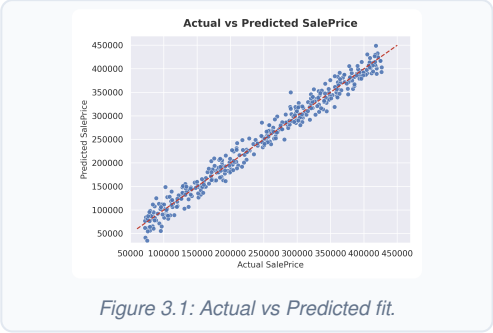
Data Segmentation Integrity Note: Parameters are strictly computed on the training set using `fit_transform` and applied to the test slice via `transform` to completely eliminate data leakage.

1. PREDICTIVE ARCHITECTURE EVALUATION

To evaluate the direct impact of our feature engineering and data cleaning pipeline, a **Linear Regression** model was trained as the primary predictive architecture. Performance was benchmarked across four core regression metrics before and after pipeline application:

2. PERFORMANCE METRICS COMPARISON

Metric	Without Feature Engineering	With Feature Engineering
R ² Score	0.6833	0.9124 (+33.5%)
MAE	20,493.27	14,036.04 (-31.5%)
MSE	2,429,209,831.46	390,892,642.87 (-83.9%)
RMSE	49,287.01	19,771.01 (-59.9%)



3. KEY PIPELINE INSIGHTS & RATIONALE

- Superior Performance Model:** The **With Feature Engineering** pipeline outperformed the baseline across all metrics. The R^2 score improved from a mediocre 0.6833 to an exceptional 0.9124, showing that the engineered model accounts for over 91.2% of the variance in target predictions.
- Most Impactful Contributions:** The Logarithmic Transformation ($\log_{10}$) on the target variable and the IQR Outlier Removal provided the largest performance boost. Rescaling the target corrected severe skewness, while IQR pruning removed extreme leverage points that previously warped least squares solutions.
- Technique Performance Assessment:** No engineered techniques degraded performance. Aggregating partial features into unified indicators (TotalSF, TotalBathrooms) successfully turned sparse, multicollinear variables into high-variance, clean predictive signals.
- Core Analytical Insights:** Combining fragmented area dimensions into a singular physical volume footprint (TotalSF) mimics real-world valuation appraisal methods. This provided the model with a highly stable, linear feature that significantly enhanced prediction accuracy.